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How does digital transformation enhance enterprise technological innovation? Evidence from Chinese manufacturing listed companies

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ABSTRACT

Using the data of listed companies in the Shanghai and Shenzhen A-share manufacturing industry from 2013 to 2020, this paper empirically tests the effect and mechanism of digital transformation (DT) on enterprise technological innovation (ETI). The results show that DT can significantly promote ETI, and there is an inverted-U relationship between DT and ETI. DT can improve the ETI through the demand scale effect, collaborative innovation effect, and knowledge spillover effect. And the above effects explain the inverted-U relationship between DT and ETI. This paper further explores the role of digital infrastructure (DI) and finds that DI can enhance the facilitating role of DT and ETI. Moreover, demand scale effect and collaborative innovation effect work only at high-level DI regions. This article provides enlightenment for promoting enterprise DT and ETI.

1. Introduction

Driven by the new generation of digital technology, the degree of digital transformation (DT) of enterprises is constantly improving. Production, supply, research and development, consumption, and other links are effectively connected. DT represents a dynamic transformation process where the ongoing application of digital technologies reshapes enterprise production, operations, and management. It focuses on combining information, computing, communication, and connectivity technologies. DT is not only a technological transformation, but also a more complex process that requires the full utilization of all resources (Gong & Ribiere, 2021).To continuously break through and replace existing rules, DT reshapes or disrupts innovation processes and models (Hinings et al., 2018; Verhoef et al., 2021; Vial, 2019). Some scholars have defined DT from the perspective of its influence. Digital technology not only digitizes resources but also creates value and revenue from digital assets (White, 2012). DT is the integration of multiple new digital technologies, transforming multiple business dimensions, including business models, operations, etc., to influence employee skills and corporate culture, achieving excellent performance and sustainable competitive advantage. DT alters, complements, or replaces existing game rules within organizations and fields, with profound implications for innovative organizational forms. Moreover, DT is not limited to startups and tech giants, but embraces companies of all sizes (Bresciani et al., 2021). The impact of DT on enterprises is comprehensive. Technological innovation is an important means for enterprises to gain competitive advantages. Can DT also influence technological innovation?

Technological innovation refers to the innovation of production technology, including the development of new technologies or the application of existing technologies for innovation. Existing research has explored the driving factors of technological innovation, including market demand, government subsidies, property rights protection, and knowledge spillovers (Chen et al., 2024; Petersen et al., 2016; Wang & Seidle, 2017; Yang & Wang, 2024). DT can also influence the above factors. The digital economy has greatly reduced market friction, especially the reduction of search, entry, and transportation costs, which have a profound impact on the innovation value of enterprises (Chen, 2020). DT is increasingly recognized as a means of promoting innovation, providing greater flexibility in identifying emerging trends or innovations, making decisions, and responding to changes more quickly (Tajudeen et al., 2022). Moreover, DT is a comprehensive process that not only adopts new technologies, but also requires organizations to achieve a strategic shift, which leads to a change in innovation models (Vărzaru & Bocean, 2024). Research has found that Chinese companies are facing innovation dilemmas. DT can overcome innovation challenges by improving innovation quality and enhancing absorption and transformation capabilities (Zhuo & Chen, 2023). Therefore, in the context of vigorously developing the digital economy, it is of great theoretical and practical significance to study the effect and mechanism of DT on the technological innovation of enterprises (ETI).

This paper mainly studies the impact of DT on ETI. We first define

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and measure DT and ETI. The current research recognizes that DT is a transformation and upgrading process in which an enterprise uses digital technology to combine with production and business to change its original business model, organizational structure, management model, decision-making model, and supply chain collaboration model. DT can not only create new consumption and supply mechanisms (Wong et al., 2015), improve the matching accuracy of supply and demand (Pisano et al., 2015), but also promote the transformation and upgrading of primary traditional industries. To facilitate understanding and measurement, this paper argues that the DT of enterprises is a process of internal and external transformation containing business processes, organizational structure, business model, and culture, which supported by digital technologies, digital products, digital investments, and digital infrastructure. Since there is no open, unified data source to help us measure DT, scholars mainly take the text analysis approach (Li et al., 2023; Yang et al., 2023). However, text analysis methods, especially dictionary methods, are not accurate in measuring DT. It cannot recognize the negative meaning of the sentence and overestimates the degree of DT. So, we establish a digital transformation index to measure the performance of enterprises in digital strategy, digital investments, digital R&D, and digital technology applications. The scope of ETI is also relatively broad. According to data availability, we measure enterprise innovation from a patent perspective. Specifically, three types of patents, namely invention, utility model, and exterior design, are used to measure ETI. Then, this paper uses the fixed effects model to explore the average impact of DT on ETI. Given the properties of DT, we use a quantile regression model to further test whether DT has a non-linear effect on ETI. We analyze the theoretical mechanism of DT's impact on ETI from the dimensions of demand scale, collaborative innovation, and knowledge spillover. Finally, we further analyze the role of DI and find it can strengthen the promotion effect of DT on ETI.

The marginal contributions of this paper are as follows: First, we construct the digital transformation index. Existing research mainly uses objective data methods and text analysis to measure DT. Acemoglu and Restrepo (Acemoglu & Restrepo, 2019) investigated the impact of robots on labor markets. Brynjolfsson et al. (Brynjolfsson et al., 2021) found that factories with predictive analysis had higher productivity. However, the number of robots and the predictive ability of enterprise data can only reflect one aspect of DT, not comprehensively. Although text analysis methods can comprehensively obtain keywords related to DT in the annual reports of companies by setting up dictionaries related to DT, they cannot effectively identify the negative meaning of sentences. Previous studies have used the dictionary method to measure the DT of listed companies and analyzed the impact of DT on green innovation, business performance, sustainable development, and other aspects of enterprises (Du et al., 2024; Zhai et al., 2022). Practically, the impact of DT may be overestimated. Therefore, we construct a digital transformation index from four aspects and it can comprehensively reflect the level of digital transformation of enterprises. Secondly, existing research mainly investigates the average impact of DT on innovation (Zhang et al., 2024). We use the quantile regression model to investigate the non-linear relationship between DT and ETI. And we find an inverted-U-shaped relationship between DT and ETI. Finally, existing research mainly focuses on the mechanism from internal perspectives, such as financing constraints and agency conflict (Liao et al., 2023), or external perspectives, including market competition and knowledge spillover (Gu et al., 2023; Wei et al., 2024). This paper not only considers the demand scale effect but also analyzes the external mechanism role of inter-firm collaborative innovation and knowledge spillover. This paper complements the existing literature on the mechanism of DT and enterprise innovation.

2. Literature review and theoretical mechanism

2.1. DT and enterprise innovation

Most literature believes that DT can promote ETI. DT brings transformative digital functions to enterprises, which can stimulate enterprise innovation (Autio et al., 2018). DT supports the company's performance with lower cost and higher operational efficiency, brings long-term performance to the enterprise (Zhai et al., 2022). DT also has an important role towards new product development (Dabić et al., 2023). The boundaries of the digital innovation process are broken, and the innovation output is increased. Digital management has endowed enterprises with innovation capabilities, which will form positive feedback regarding innovation output (Rachinger et al., 2019). Wen et al. (Wen et al., 2022) analyzed the relationship between DT and enterprise investment from the perspective of enterprise competitive strategy. They found that in the process of DT of the manufacturing industry, enterprises' input in innovation activities increased significantly. Moreover, as a critical factor, DT could enhance the innovation performance of enterprises together with other factors, such as enterprise scale, idle resources, and network embedment (Xing et al., 2023). Moreover, DT not only transform existing innovation activities but also form innovation combinations and bring competitive advantages to enterprises (Zhang et al., 2023). The emergence of novel and powerful digital technologies, digital platforms, and digital infrastructure has changed innovation to a large extent, not only providing new opportunities for innovators but also having a broad impact on value creation and value capture (Nambisan et al., 2019).

DT may have an inverted U-shaped impact on ETI. According to the theory of network externalities and the law of diminishing marginal returns on factors in economics, we can infer the existence of an inverted U-shaped relationship between the two. The theory of network externalities is the source of economies of scale on the demand side. When the value of a product to consumers increases with the number of other users, the product has network externalities. With the continuous integration of enterprises into digital transformation platforms or networks, the range of transactions and interactions that enterprises can engage in increases. The law of diminishing marginal returns on factors means that as the input of digital transformation factors continues to increase, the output or benefits per unit factor initially increase, but after reaching the optimal point, the output brought by digital transformation factors would become smaller and smaller.

DT enables enterprises, universities, research institutes, and other entities to form an open innovation system, and the "network effect" have the characteristics of increasing effect. At this point, the impact of DT on ETI is still in the upward phase of the inverted U-shaped curve. DT requires a significant capital investment and faces strong resource constraints initially. Therefore, continuously increasing investment in digital elements could lead to an increase in innovation performance. With the increase of the degree of DT and information density, various marginal costs of enterprises could decrease, the cost of innovation decreases, and the efficiency of innovation output increases. Under the law of diminishing marginal returns, the continuous increase of digital factor inputs can lead to the decline of unit investment utility (Wamba et al., 2017). Digital investment can also have a crowding effect on other R&D investments of enterprises. If the digital investment is within the optimal allot efficiency level, resource misallocation may appear, resulting in idle digital resources or a relative shortage of other resources. According to the IT productivity paradox proposed by Solow, digital technology may have adverse or insignificant effects on enterprise innovation (Hajli et al., 2015). While bringing benefits to enterprises, DT could also generate high management costs (Ekata, 2012). Moreover, DT is a long-term process, and the mismatch between the existing organization, management model, and the technical framework of DT may lead to chaos, increase the adaptability cost of enterprises, and inhibit enterprise innovation (Karhade & Dong, 2021). In addition, excessive digital

investment may lead to information overload, forcing enterprises to reduce the breadth and depth of knowledge utilization (Dong & Netten, 2017). At this time, further increases in digital investment do not improve the marginal output of innovation, and the driving effect on innovation activities gradually declines until there is no promotion effect. From the above analysis, we conclude that there is an inverted-U-shaped relationship between DT and enterprise innovation.

Existing researchers have yet to form a consensus view on the relationship between DT and ETI. On the one hand, different scholars have various definitions of innovation, and the measurement of enterprise innovation from the aspects of innovation organization mode, process innovation, product innovation, technological innovation, or other aspects may lead to significant differences, resulting in different impacts of DT on innovation. On the other hand, the stage of DT of enterprises in different countries is various. The United States and Germany adopt policies to promote DT early, and the degree of DT ranks among the highest in the world. However, in some developing countries, the level of development of the digital economy and DT is slow. Therefore, the conclusions on the impact of enterprises' DT on innovation may be different in different countries. Based on the network externalities and the law of diminishing marginal returns on factors in economics, this paper proposes the following hypothesis:

H1. DT can significantly promote ETI and has an inverted-U-shaped feature.

2.2. Theoretical mechanism

Based on the demand-pull theory, open innovation theory and knowledge spillover theory, we analyze the mechanism by which DT affects ETI from the dimensions of market demand, collaborative innovation, and knowledge spillover (see Fig. 1). Analyzing these theocratical mechanisms is conducive to opening the "black box" of DT on ETI.

2.2.1. Demand scale effect

According to the demand-pull theory, market demand is the main driving force of technological innovation (Schmookler, 1966). Market demand provides a clear direction for the development of enterprises and reduces the uncertainty of innovation. The high profits created by market demand motivate companies to innovate to gain a competitive advantage. Consumers also play a role of feedback and participation in the innovation process. Government procurement is a typical way of demand-driven innovation (Liu et al., 2024), which can reduce the risk of supplier innovation (Dalpé, 1994). Dynamic capability theory argues that firms are able to integrate, build, and reallocate internal and external resources in a rapidly changing environment to adapt to market changes and remain competitive. In the context of DT, the dynamic capability theory can well explain the important mechanism of market demand. On the one hand, the data capture and analysis tools provided by DT enhance the perception ability of enterprises, enabling enterprises to quickly identify changes in market demand (Wijayarathne et al., 2024). On the other hand, DT also improves the ability of enterprises to reallocate resources, which makes it convenient for enterprises to devote limited resources to innovation.

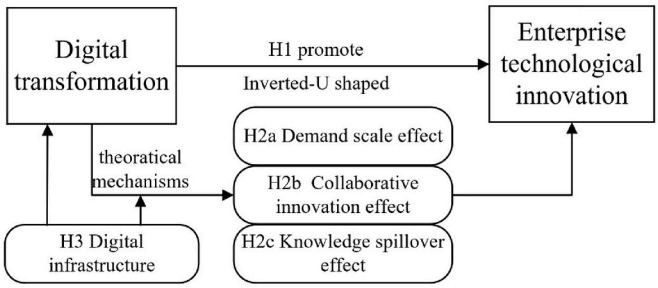


Fig. 1. Theoretical mechanism analysis.

DT has not only changed the way enterprises collect market demand information but more profoundly, it has also changed the model of market demand-induced innovation. DT provides channels for enterprises to collect demand-side information (Zhai et al., 2022), thus enhancing the signal guidance role of the demand market for enterprise innovation. DT can effectively reduce consumer search costs, expand market scope, and improve market matching efficiency (Goldfarb & Tucker, 2019), enabling enterprises to obtain more consumer purchase information at a lower cost. More personalized consumer demand and business model innovation can provide references for enterprises' technological innovation (Lee, 2013). Enterprises undergoing DT can make full use of the advantages of demand scale and tap the "big data" of demand, which helps enterprises open up new markets and develop new products and new technologies. Furthermore, DT has changed traditional innovation processes and models. Decentralization, digitization, and open source have become popular trends in the industrial sector. Malewska et al. (Malewska et al., 2024) found that under the drive of digital technology, enterprises expanded new markets by innovating their business models. Nambisan (Nambisan, 2002) pointed out that customers can participate in value-creation activities such as product design and development, product testing, and product marketing through virtual customer environments. DT has blurred the boundary between process innovation and product innovation (Vick et al., 2015) and also has brought many derivative innovations (Nylén & Holmström, 2019). We pose the hypothesis H2a:

H2a. DT can promote technological innovation through the demand scale effect.

2.2.2. Collaborative innovation effect

The increase in technological complexity makes single innovation entities or linear innovation organizational models unsuitable for complex innovation development. The collaborative innovation organizational model has become the main form of innovation today. Open innovation theory emphasizes that enterprises should break organizational boundaries in the innovation process and make full use of internal and external resources and knowledge to accelerate innovation. It also advocates that companies innovate more efficiently and flexibly by sharing knowledge, technology, and resources with external partners. According to the theory of open innovation, collaborative innovation among enterprises can integrate internal and external resources, reduce research and development costs, and improve market adaptability. It not only expands business opportunities for enterprises, but also accelerates the innovation process. Yuan and Li (Yuan & Li, 2024) concluded that the collaborative model of the innovation ecosystem enhanced the innovation and survival capabilities of enterprises by promoting cooperation with other entities within the system. In the context of DT, open innovation theory provides an important perspective for understanding the mechanism of collaborative innovation. DT lowers the barrier to knowledge sharing and collaboration through digital technologies (such as cloud computing, big data, artificial intelligence), enabling enterprises to integrate internal and external resources more efficiently. Moreover, DT promotes the formation of an innovation ecosystem, where companies develop new technologies with external partners through a platform-based strategy. For example, tech giants such as Microsoft and Google collaborate with developer communities through open APIs (application programming interfaces) to drive innovation in software and services (West & Gallagher, 2006).

DT enabled the implementation of new forms of collaboration between firms (Rachinger et al., 2019). DT provides a platform and tools for collaborative innovation in enterprises and promotes the dissemination of collaborative innovation culture. The broad penetration advantage of DT can break through the internal innovation boundary of enterprises and enhance the ability of collaborative innovation of enterprises. Digital platforms supported by digital technologies are an

essential form of innovation cooperation with different enterprises, individuals, research institutes, universities, and other entities. Gong and Li (Gong & Li, 2023) found that the government provided boundary resources to users through open digital platforms, enabling users to achieve innovation complementarity. Strategic alliances are a popular form of organization for biotech companies to collaborate on external knowledge acquisition (Subramanian et al., 2018). Tian and Lu (Tian & Lu, 2023) found that digital infrastructure construction encouraged enterprises to innovate collaboratively across regions. More importantly, DT has impacted the organizational culture of enterprises. Vick et al. (Vick et al., 2015) empirically validated the typology of information culture and found that different cultures had an impact on knowledge creation and technological innovation. Butt et al. (Butt et al., 2024) found that the organizational culture of leading enterprises is strategic and cutting-edge. DT accelerates the dissemination speed of information and culture among different subjects, promoting the spillover of collaborative innovation culture to other subjects. Malewska et al. (Malewska et al., 2024) showed that DT indirectly influenced the business model innovation of energy enterprises through digital organizational culture. Therefore, we pose the hypothesis H2b:

H2b. DT can promote collaborative innovation to improve the ETI.

2.2.3. Knowledge spillover effect

As a public good, knowledge is non-competitive and non-exclusive, able to flow freely between different subjects and generate externalities. Knowledge spillovers usually occur between firms, research institutions, and individuals, and are disseminated through formal or informal channels, thereby promoting innovation and economic growth (Arrow, 1972; Romer, 1986). Griliches (Griliches, 1992) further noted that knowledge spillovers could be achieved through a variety of avenues, such as R&D collaborations, personnel mobility, patent citations, and geographical proximity. The core idea of knowledge spillover theory is that the dissemination and sharing of knowledge can improve the efficiency of innovation in society as a whole, although knowledge creators may not be able to fully capture its benefits. The main motivation for innovative enterprise cooperation is to acquire knowledge resources, and knowledge spillovers can affect the innovation performance of enterprises (Bernal et al., 2022). Knowledge spillover comes from the constantly updated knowledge stock under experiential learning. Different from the replication function of knowledge transmission, knowledge spillover is the re-creation and innovation of knowledge, which can gather the wisdom of innovative subjects and continue to benefit participants. Choi and Lee (Choi & Lee, 2021) used data from American manufacturing companies and concluded that knowledge spillovers had a positive moderating effect on technological diversification and R&D productivity. Colombelli et al. (Colombelli et al., 2024) used data from Italian startups and found that digital knowledge spillovers provided new opportunities for the development of innovation and entrepreneurship.

DT has changed the way knowledge exchanged and increased the rate of knowledge spillovers. Knowledge spillover theory believes that entrepreneurs are the key channel for knowledge dissemination, and the creation of knowledge depends on different learning and interactive processes (Audretsch & Keilbach, 2007). The geographical proximity of actors is a necessary factor in generating knowledge spillovers (Lee et al., 2013). However, in the context of DT, knowledge spillover can break through the limitations of time and space. Zhang et al. (Zhang et al., 2024) empirically analyzed the impact of manufacturing intelligence on technological innovation and found that intelligence had horizontal and vertical spillover effects on technological innovation, and a positive impact on technological innovation in related industries. Yang et al. (Yang et al., 2023) found that DT had a positive spatial spillover effect on the innovation of neighboring cities. Du et al. (Du et al., 2024) also concluded that DT generated local proximity effects, which had an impact on the innovation of surrounding enterprises. Furthermore, DT facilitates the explicit use of tacit knowledge, enabling experiences and skills that would otherwise be difficult to disseminate widely through digital tools (e.g., online training, virtual reality). Thus, we pose the hypothesis H2c:

H2c. DT can promote ETI through knowledge spillovers.

2.3. DT and digital infrastructure

DT is intrinsically linked to the development of digital infrastructure (DI). Existing studies have extensively analyzed the concept of DI. Yoo et al. (Yoo et al., 2010) defined DI as a modular and reusable technological platform that facilitates innovation and collaboration across organizations and industries. Similarly, Henfridsson et al. (Henfridsson et al., 2014) characterized DI as a complex system comprising hardware, software, network protocols, and data standards, which provides foundational technical capabilities to support DT for both society and organizations. Monteiro et al. (Monteiro et al., 2013) further conceptualized DI as an ecosystem integrating technological components, social norms, and institutional arrangements, serving as a driving force for DT by enabling cross-organizational and cross-geographical collaboration and innovation. Based on the above definition, this paper believes that DI includes hardware, software, networks, and data storage, which can play a key role in promoting collaboration and advancing DT.

There exists a profound and symbiotic relationship between DT and DI, where DI serves as the technological backbone for DT, while the evolving demands of DT propel the continuous advancement and expansion of DI. Firstly, DI provides essential technological support for DT. Information technology infrastructure serves as the primary driving force for the adoption of digital technology (Omrani et al., 2024). The rapid proliferation of advanced technologies such as big data analytics, cloud computing, and artificial intelligence has empowered businesses to process vast amounts of data with unprecedented efficiency, thereby optimizing decision-making processes. Furthermore, the advent of edge computing and the deployment of 5G networks have significantly addressed DT's requirements for low latency and high bandwidth, thereby facilitating the development of smart manufacturing and smart cities (Shi et al., 2016). Additionally, the inherent openness and scalability of DI create a fertile environment for innovation within DT. For instance, the emergence of the platform economy has further catalyzed the evolution of DI, transforming it into a pivotal platform that connects supply and demand, fostering resource sharing and value co-creation. Secondly, the social and technological embeddedness of DI offers institutional safeguards for DT. DI encompasses not only technological components but also social norms and institutional frameworks, such as the establishment of data privacy and security standards. These frameworks provide a compliance structure that enhances consumer trust in digital services, thereby supporting the sustainable growth of corporate DT initiatives (Zuboff, 2019). Lastly, the dynamic interplay between DI and DT forms a virtuous cycle of mutual enhancement. As the demand for DT continues to escalate, traditional infrastructure often falls short in meeting the requirements for flexibility, scalability, and security. For example, the integration of blockchain technology has introduced robust technical conditions for ensuring data security and transaction transparency, thereby further advancing DT applications in critical sectors such as finance and supply chain management. In summary, the interdependence between DI and DT is not only fundamental but also dynamic, driving continuous innovation and societal progress in the digital era. Thus, we pose the hypothesis **H3**:

H3. High-level DI plays a pivotal role in accelerating and enhancing the process of DT and it may also affect the mechanism of DT on ETI.

3. Research and design

3.1. Data

This paper uses the data of Shanghai and Shenzhen A-share manufacturing listed companies from 2013 to 2020. Patents and financial data are from the China Stock Market & Accounting Research Database (CSMAR) database, DT data are from the Chinese Research Data Services Platform (CNRDS) database. Relevant mechanism variables are from the CNRDS database and China Statistical Yearbook. The classification of the manufacturing industry refers to the Guidance on Industry Classification of Listed Companies (2012 Revision version). Based on the existing research, some data are excluded according to the following criteria: (1) enterprises delisted from 2013 to 2020; (2) firms with missing financial information. We winsorize the continuous variables at the 1 % and 99 % levels. Finally, we obtained a total number of 10,488 baseline regression sample data.

3.2. Description of variables

3.2.1. Explained variables

Most literature measures the technological innovation of enterprises (ETI) from the perspective of different types of patents (Bernal et al., 2022). Some literature measures ETI using the quantity and quality of innovation input and innovation output. In China, there is usually a two-year delay in patent authorization. Considering the long patent grant cycle, this paper uses the number of patent applications (*Patent*1) and the number of invention patent applications (*Patent*2) as the explained variables. Considering that the number of patent applications of some enterprises may be skewed to the right, we use the natural logarithm of the number of patent applications increased by one as the proxy variable of ETI. Table 2 shows that the difference in *Patent*2 among enterprises is greater than *Patent*1. This also indicates that invention patents can better reflect the quality of technological innovation.

Table 1
Digital transformation index.

Dimensions	Sub-dimensions	Indicator
1 Digital strategy	1a Creation of digital roles in management	1a1 Set up a chief information officer (CIO) (Yes, 1; No, 0) 1a2 Set up a chief technology officer (CTO) (Yes, 1; No, 0) 1a3 Set up a chief digital officer (CDO) (Yes, 1; No, 0)
2 Digital investments	2a Digital capital investment	2a1 Number of digital projects and project amount invested
	2b digital human capital investment	2b1 Number of hires for digital positions
	2c Construction of digital innovation base	2c1 Number of national science and technology innovation bases firms participating
3 Digital R&D	3a Digital basic research	3a1 Number of national and industry standards firms participating 3a2 Number of digital papers published by companies
	3b Digital application research	3b1 Number of types of digital innovation projects 3b2 Number of digital-related national science and technology projects
4 Digital technology application	4a Artificial intelligence technology	4a1 Frequency of words of AI technology
	4b Blockchain technology	4b1 Frequency of words of blockchain technology
	4c Cloud computing technology	4c1 Frequency of words of cloud computing technology
	4d Big data technology	4d1 Frequency of words of big data technology

Notes: The DTI refers to the CSMAR database and can be seen at https://data.csmar.com/. The application indicators of digital technology adopt the method of word frequency statistics. To improve the accuracy of the data, we have removed expressions with negative words before keywords.

3.2.2. Explanatory variable

There are usually four methods for measuring DT, including text analysis, index construction, selecting specific variables, and constructing event shocks. Gal et al. (Gal et al., 2019) and Zhai et al. (Zhai et al., 2022) quantify DT from digital technology applications and business model innovation. They used word frequency to capture the degree of enterprise DT. But text analysis method cannot identify the actual implementation of DT. The scope of DT is extensive, and there is no special department in China to collect data on DT. Gopalan et al. (Gopalan et al., 2022) consider the use of a company's website as a proxy variable for DI. However, the above measurement methods only reflect one aspect of DT, which is not conducive to reflecting the comprehensive impact of DT. So, some scholars use index construction methods to comprehensively demonstrate the performance of DT. Skare et al. (Skare et al., 2023) use the official European database to construct a digital transformation index (DTI) from the perspectives of human capital, connectivity, integration of digital technology, and digital public services. To comprehensively measure DT, we refer to the index construction method from the CSMAR website to construct the DTI from four dimensions, including digital strategy, digital investments, digital R&D, and digital technology applications (see Table 1). We use the entropy method to construct DTI. The specific steps are as follows: (1) Standardize the indicators. (2) Calculate the proportion of indicators. (3) Calculate the entropy value of the indicator. (4) Calculate the coefficient of difference for the indicators. (5) Empower indicators. (6) Calculate sample evaluation values through weights. Finally, we standardize the DTI in units of 100. Table 2 shows that significant differences are in DT among enterprises, with most of them being at a junior level.

3.2.3. Mechanism variables

(1) Demand scale effect. Generally, enterprises with enormous operating profits have more market demand shares. They can better mine consumer information and occupy the advantage of demand scale in the process of DT. Therefore, *OE_profit* is the proxy variable of enterprise demand scale advantage. In addition, the exploitable demand advantage of enterprise DT is also reflected in the fact that enterprises can obtain consumer demand evaluation at a lower cost. More diversified digital collection methods help enterprises collect users' preferences. Therefore, whether the enterprise conducts a customer satisfaction survey (*Client*) is the supplementary mechanism variable.

(2) Collaborative innovation effect. Enterprise collaborative innovation is reflected in the way of multi-agent cooperation in innovation strategy, and the innovation results. Therefore, according to the enterprise annual report, we collect the words related to cooperation, such as cooperation, win-win, communication, etc., and use the proportion of the word frequency as the proxy variable of enterprise cooperation culture. Moreover, according to Yang et al. (Yang et al., 2023) and Tian and Lu (Tian & Lu, 2023), we also use jointly applied patents to reflect their collaborative innovation achievements. Joint patent application refers to a patent jointly applied for by enterprises, individuals, universities, or research institutes. It can reflect the degree of collaborative innovation between enterprises and other innovative entities.

(3) Knowledge spillover effect. In the study of knowledge spillover variables based on existing literature (Forman & van Zeebroeck, 2019), this paper uses the number of citations of patent applications(*Sum_Cite*) and the net number of citations excluding

Table 2
Descriptive statistics.

Type	Variables	Meaning	Definitions	Mean	Std.	Max.	Min.
Explanatory Variables	*DCG_index*	Digital transformation index	See details information in 3.2.2.	35.767	10.171	79.812	22.990
Explained variables	*Patent*1	Patent applications	Ln (total number of patent applications+1)	3.010	1.580	9.325	0.000
	*Patent*2	Invention patent applications	Ln (number of invention patent applications+1)	3.723	1.752	10.232	0.000
Control variables	*Size*	Firm size	Total assets of the enterprise	21.987	1.172	27.547	17.779
	Liability	Asset-liability ratio	Total liabilities/total assets	0.385	0.198	1.957	0.008
	Returns	Return on equity	Annual net profit/average balance of shareholders' equity	0.059	0.296	1.751	−14.819
	Turnovers	Asset turnover	Operating income/average total assets	0.665	0.418	8.455	0.000
	Fixed Assets	Proportion of fixed assets	Net fixed assets/total assets of enterprises	0.229	0.138	0.810	0.000
	Growth	Growth rate of operating income	(Current year's operating income/previous year's operating income) −1	0.289	4.942	2.100	−0.467
	Loss	Firm loss	If the enterprise is a loss, the Loss will be 1; otherwise, 0	0.095	0.293	1.000	0.000
	*Top*10	The shareholding ratio of the top 10 shareholders	The number of shares held by the top 10 shareholders compared to the total number of shares	0.593	0.147	1.012	0.092
	FirmAge	Firm age	Ln (difference between enterprise sample year and establishment year+1)	2.859	0.313	4.143	1.609
	Logsubsidy	Government subsidies	The logarithmic value of the total amount of government subsidies received by enterprises	15.440	2.169	22.106	5.635
Mechanism variables	*Oeprofit*	Operating profit	Annual operating profit of the enterprise	4.5702	20.1895	666.3508	−184.999
	Client	Customer satisfaction	If the firm conducts a customer satisfaction survey, Client = 1; else, 0.	0.501	0.500	1	0
	*Co_Patent*1	Joint patent applications	The natural logarithm after the total number of joint patent applications	0.675	1.138	8.367	0.000
	*Co_Patent*2	Weighted joint patent applications	The natural logarithm after the weight assigned to joint patent applications	0.934	1.434	8.969	0.000
	*Co_Patent*3	Joint invention patents	The natural logarithm after the number of joint invention patents	0.384	0.801	7.445	0.000
	Corporation	Words related to corporate culture	The logarithmic value of the total frequency of words related to corporate culture	37.957	24.420	475	0
	Sum_Cite	Cited enterprise patents	The logarithmic value of the number of cited enterprise patents	2.925	1.583	10.623	0
	Net_Cite	Net cited enterprise patents	The logarithmic value of the number of net cited enterprise patents	2.905	1.583	10.486	0

self-citations (*Net_Cite*) to represent the knowledge spillover effect of enterprises.

3.2.4. Control variables

We use the following variables controlling the business status and nature of enterprises, including company size (*Size*), asset-liability ratio (*Liability*), return on equity (*Returns*), asset turnover (*Turnovers*), proportion of fixed assets (*Fixed Assets*), growth rate of operating income (*Growth*), firm loss (*Loss*), shareholding ratio of top 10 shareholders (*TOP*10), firm age (*FirmAge*), and government subsidies (*Logsubsidy*). Detailed information about the control variables can be seen in Table 2.

3.3. Model

The explained variable in the model (1) represents the technological innovation of the firm, $Patent_{i,t}$ represents the firm i and year t. β_1 represents the causal effect of DT on technological innovation. Controls are a series of enterprise-level control variables, and this paper controls the fixed effects of year, firm, and city (θ_t、 μ_j 、 ρ_m).

$$Patent_{i,t} = \beta_0 + \beta_1 DCG_index_{it} + \sum \gamma\, Controls_{i,t} + \theta_t + \mu_j + \rho_m + \varepsilon_{i,t} \quad (1)$$

3.4. Descriptive statistical analysis

Table 2 shows the variables' descriptive statistics. We can also see the detailed definitions of these variables. The average value of DT (*DCG_index*) is 35.767, and the standard deviation is 10.171, indicating that enterprises have significant differences in DT. The minimum value of ETI (*Patent*1 and *Patent*2) is 0, showing that some enterprises have a low innovation level. We presented the data characteristics of the manufacturing industry's segmented industries in Table 3.

Based on the industry binary codes of listed companies, we have compiled and analyzed 29 manufacturing sub-sectors. And we rank

Table 3
Statistical analysis of manufacturing industry description.

Rank	Code	Name	N	*DCG_index*	*Patent*1	*Patent*2	*Size*	*Liability*	*Growth*
1	39	Manufacture of computers, communications, and other electronic equipment	382	46.512	3.608	4.405	21.869	0.362	0.268
2	38	Electrical machinery and equipment manufacturing	254	43.136	3.557	4.297	21.959	0.404	0.194
3	40	Instrument and meter manufacturing industry	58	42.762	3.072	3.792	21.069	0.252	0.172
10	37	Railway, marine, aerospace, and other transport equipment manufacturing	56	35.330	4.026	4.797	22.672	0.448	0.402
15	22	Paper and paper products industry	35	31.116	2.378	3.029	22.341	0.443	0.294
20	15	Wine, beverages, and refined tea manufacturing	43	30.571	2.251	2.650	22.364	0.324	0.157
25	27	Pharmaceutical manufacturing	229	29.207	2.310	3.039	21.933	0.303	0.462
27	32	Non-ferrous metal smelting and rolling processing industry	73	28.950	2.615	3.338	22.537	0.470	0.144
28	28	Chemical fiber manufacturing	27	28.632	2.281	2.996	22.485	0.476	0.261
29	25	Petroleum processing, coking, and nuclear fuel processing industries	23	26.353	1.571	2.087	22.426	0.502	0.079

Note: Due to space limitations, we have only presented some manufacturing industry segments.

them based on the industry's average DT. We present some of the results. The DT of the computer, communication, and other electronic equipment manufacturing industry ranks first while the petroleum processing, coking, and nuclear fuel processing industries rank the last. We can infer that the level of DT is higher in high-tech industries and lower in heavy industries. We also showcased industries ranked in other categories, with precision manufacturing at a higher level of DT, while the primary food processing industry has a lower level. The pharmaceutical manufacturing industry is also at a backward level, as most enterprises are concentrated in the industry 3.0 stage.

4. Empirical analysis

4.1. Benchmark regression results

As seen from the benchmark regression results in Table 4, the coefficient of DT on ETI is significant at the 1 % level. After adding control variables, the coefficient decreases, but it is still significant. After controlling the fixed effect of firm, city, and year, the coefficients are 0.0232 in column (3) and 0.0249 in column (6), respectively. The coefficients indicate that each increment of one unit of DT increases the ETI level by 2.32 % and 2.49 % on average. Accordingly, in response to hypothesis H1, DT can significantly promote ETI, especially for the invention patents.

4.2. Non-linear analysis

This article uses two methods to analyze the non-linear impact of DT on ETI. First, we use the quantile regression to demonstrate the trend of coefficient changes, we plotted Fig. 2. It shows the partial impact results from the 10th to 90th percentiles. When the level of enterprise technological innovation is at the initial level, the coefficient of DT on ETI is relatively high, and it reaches the highest level at the 20th percentile. As the level of ETI continues to improve, the impact of DT gradually weakens, presenting an inverted-U-shaped impact feature. Second, we add the quadratic term of *DCG_index* on the basis of benchmark regression. Based on the method proposed by Fernhaber and Patel (Fernhaber & Patel, 2012), we adopt the following strategy to test the inverted U relationship. If the quadratic coefficient is significantly negative and the abscissa value of the vertex is within the range of *DCG_index*, we can prove the inverted U-shaped relationship exists. From column of (1) of Table 5, the coefficients of *DCG_index_squ* are significantly negative, and the abscissa value of the vertex is within the range of *DCG_index*. At this point, the law of diminishing marginal returns on factors exceeds the cost-cutting effect brought about by DT. Continuously increasing investment in DT does not necessarily lead to sustained technological innovation. We can verify hypothesis H1 that there is an inverted U-shaped relationship between DT and ETI. The mechanism analysis in the following paper also verifies this conclusion.

4.3. Robustness test

This paper conducts the tests in the following ways to ensure the robustness of the conclusion on the impact of DT on ETI. First, we change the explained variable. We use the natural logarithm of the total number of patent grants (*Award*1), the natural logarithm of the number of invention patent applications (*Award*2), and the intensity of R&D investment (*RD*1) of the firm (expressed by the ratio of the annual R&D investment of the firm to the total assets) respectively as the new explained variables. Table 5 of columns (2)–(4) indicates that DT can significantly promote ETI.

Second, we use the new explanatory variable of DT. We use DT's secondary indicators, digital investment, and digital R&D, as the new explanatory variables. DT relies on the investment of material and human capital by enterprises. Moreover, high-level DT requires a high level of digital R&D from enterprises. So, both digital investment and digital R&D level can reflect the actual implementation status of DT. As shown in columns (5)–(8), the coefficients are all significantly positive at the 1 % level.

4.4. Endogeneity test

An apparent reverse causality exists between DT and ETI, leading to the inaccuracy of the above-estimated results. To alleviate endogeneity, we use urban topographic relief as an instrumental variable (IV) (Nunn, & Qian, 2014). Generally, the construction of digital infrastructure is less complicated in flat terrain, and infrastructure development is relatively rapid, which can promote the level of DT of local enterprises. Meanwhile, there is no direct relationship between the degree of terrain flatness and the level of ETI. The IV can only influence ETI by DT. However, topographic relief is a time-invariant variable during the sample period. We use the topographic relief of each city and the number of industrial enterprises as the interactive term as the final IV. We use the two-stage least squares method (2SLS). The first regression result is shown in the column of Table 6. The coefficient between IV and enterprises' DT is significantly negative at the 1 % level, indicating that rugged terrain is not conducive to enterprises' DT. Moreover, according to the Cragg-Donald Wald F statistics, the IVs are not weak instrumental variables. In the second stage of 2SLS, shown in columns (2) and (3), the influence of DT on ETI is significantly increased compared to the baseline results.

4.5. Mechanism analysis

To examine the three mechanisms by which DT affects ETI, we established models (2), (3) and (4). In model (2), the coefficient α_1 represents the influence of DT on the mechanism variables. In model (3), δ_2 represents the impact of the mechanism variable on ETI. If α_1 and δ_2 are both significantly positive, it indicates that DT can promote ETI through $M_{i,t}$. Based on the above analysis, we know that there is an inverted U-shaped relationship between DT and ETI. We refer to the method of Fernhaber and Patel (Fernhaber & Patel, 2012), and use model (4) to explain the role of the three mechanisms in the nonlinear relationship. φ_2 is the quadratic coefficient of DT. The existing literature mainly analyzes the movement of the vertices of the curve and the change of curvature. First, the abscissa of the vertex of curvature is $x = -\frac{\varphi_1+M\varphi_3}{2\varphi_2+2M\varphi_4}$. To understand how $M_{i,t}$ affects the abscissa change of the vertex, we calculate the first derivative of M. $\frac{\partial x}{\partial M} = \frac{\varphi_1\varphi_4-\varphi_2\varphi_3}{2(\varphi_2+M\varphi_4)^2}$. If $\varphi_1\varphi_4 - \varphi_2\varphi_3 > 0$, the vertex abscissa shifts to the right as M increases. This indicates that the improvement of M enhances the positive promotion effect of DT on ETI. Second, we focus on φ_4. If φ_4 is significantly positive, the curvature becomes larger and the curve becomes steeper as M increases. It indicates that M can strengthen the nonlinear relationship between DT and ETI. θ_t, μ_j , ρ_m, $\varepsilon_{i,t}$ and $Controls_{i,t}$ are the same meaning as in model (1).

$$M_{i,t} = \alpha_0 + \alpha_1 DCG_index_{it} + \sum \gamma\ Controls_{i,t} + \theta_t + \mu_j + \rho_m + \varepsilon_{i,t} \quad (2)$$

$$Patent_{i,t} = \delta_0 + \delta_1 DCG_index_{it} + \delta_2 M_{it} + \sum \gamma Controls_{i,t} + \theta_t + \mu_j + \rho_m + \varepsilon_{i,t} \quad (3)$$

$$Patent_{i,t} = \varphi_0 + \varphi_1 DCG_index_{it} + \varphi_2 DCG_index_squ_{it} + \varphi_3 DCG_index_{it} \times M_{i,t} + \varphi_4 DCG_index_squ_{it} \times M_{i,t} + \varphi_4 M_{i,t} + \sum \gamma\ Controls_{i,t} + \theta_t + \mu_j + \rho_m + \varepsilon_{i,t} \quad (4)$$

Enterprises with large demand scales can utilize the technological advantages of DT and guide the direction of technological innovation by collecting and analyzing consumer information. From Table 7 of

Table 4
Baseline regression results.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
	*Patent*1	*Patent*1	*Patent*1	*Patent*2	*Patent*2	*Patent*2
DCG_index	0.0613***	0.0472***	0.0232***	0.0665***	0.0510***	0.0249***
	(46.3995)	(37.3827)	(14.7266)	(46.1964)	(36.4606)	(14.3228)
Size		0.5255***	0.5899***		0.5685***	0.6308***
		(37.2575)	(40.7135)		(35.9655)	(38.6776)
Liability		0.2497***	−0.0296		0.2247**	−0.0601
		(2.9923)	(-0.3630)		(2.3428)	(-0.6425)
Returns		0.1302***	0.0933**		0.1645***	0.1201**
		(2.9924)	(2.1951)		(3.3559)	(2.4192)
Turnovers		0.1729***	0.2771***		0.1380***	0.2524***
		(5.1193)	(7.2918)		(3.6970)	(5.9646)
Fixed Assets		−1.0992***	−0.3850***		−1.1800***	−0.4347***
		(-10.3485)	(-3.5144)		(-9.6548)	(-3.4609)
Growth		0.0016	0.0012		0.0018	0.0014
		(0.5471)	(0.5606)		(0.6359)	(0.6861)
Loss		−0.2433***	−0.1366***		−0.2878***	−0.1647***
		(-4.9633)	(-3.1172)		(-5.1405)	(-3.2732)
*Top*10		0.3413***	−0.0372		0.2317**	−0.1182
		(3.7291)	(-0.4198)		(2.2517)	(-1.1780)
FirmAge		−0.1148***	−0.1224***		−0.1519***	−0.1506***
		(-2.8147)	(-2.9488)		(-3.3197)	(-3.2356)
Logsubsidy		0.0477***	0.0548***		0.0533***	0.0628***
		(7.2853)	(7.6469)		(7.3308)	(7.8048)
Constant	0.8167***	−10.7944***	−11.3480***	1.3456***	−11.0224***	−11.5511***
	(16.7090)	(-36.7217)	(-38.9414)	(24.8138)	(-33.6860)	(-35.3996)
Firm fixed effect	No	No	Yes	No	No	Yes
Year fixed effect	No	No	Yes	No	No	Yes
City fixed effect	No	No	Yes	No	No	Yes
Observations	10,488	10,488	10,488	10,488	10,488	10,488
R-squared	0.1558	0.3428	0.5198	0.1489	0.3239	0.5031

Note: Robust t-statistics are in parentheses. ***, ** and * indicate statistically significant at 1 %, 5 % and 10 % respectively.

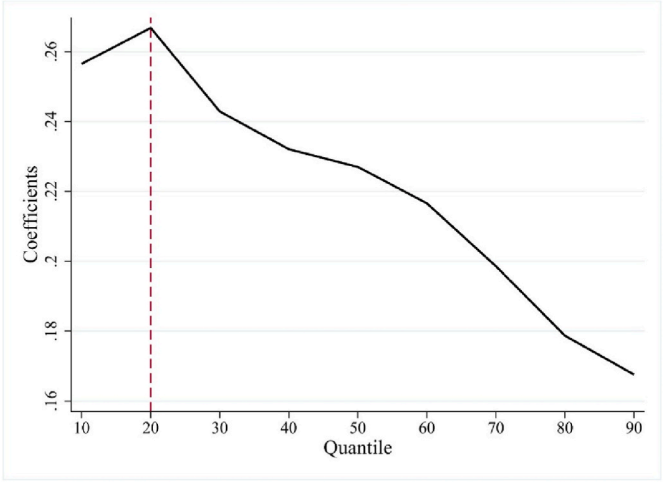


Fig. 2. Quantile regression of DT on ETI.
Note: The horizontal axis represents different quantiles. All coefficients are significant at the 1 % level.

columns (1) and (3), DT improves the operating income of enterprises and the proportion of conducting customer surveys. The coefficients of the mechanism variables in columns (2) and (4) are significantly positive at the 5 % level, indicating that the demand scale effect also promotes ETI. To account for the nonlinear relationship between DT and ETI, we substitute *OEprofit* and *Client* into model (4) separately, and finally we obtain columns (5) and (6). We calculate the effect of increasing demand scales on the abscissa of the vertices. We find that $\varphi_1\varphi_4 - \varphi_2\varphi_3 > 0$.The increase of demand scale effect promotes the positive effect of DT on ETI. Moreover, the coefficient of *DCG_squ_M* is greater than 0, and the scale effect of demand strengthens the nonlinear relationship between DT and ETI. Therefore, we can respond to hypothesis H2a that DT can improve ETI by making full use of the demand scale effect.

We use the variables of corporate cooperation culture and collaborative patent applications to measure collaborative innovation in enterprises. As can be seen in Table 8, columns (1) and (2) show that DT can improve enterprises' cooperation consciousness and cooperation culture. And cooperation can significantly improve ETI. Moreover, the cooperation consciousness among enterprises could significantly promote ETI. Furthermore, we use the total number of joint applications for the three kinds of patents (*Co_Patent*1), the natural logarithm after the weight assigned to the three kinds of joint patent applications[1] (*Co_Patent*2), and the natural logarithm after the number of invention patents (*Co_Patent*3) applications to measure collaborative innovation. From the perspective of innovation achievements, DT significantly promotes collaborative innovation at the 1 % level. We investigate the role of cooperative culture in the relationship between DT and ETI based on model (4). The results can be seen in column (6). We find that $\frac{\partial x}{\partial M} > 0$, which means that the increasing development of cooperative culture could deepen the positive effect of DT on ETI. Furthermore, the coefficient of *DCG_squ_M* is significantly positive at the 1 % level. The cooperative culture can also strengthen the nonlinear relationship between DT and ETI. Therefore, we can respond to hypothesis H2b that DT can improve the ETI by promoting collaborative innovation.

We measure the level of knowledge spillover by the number of patent citations and investigate whether DT has promoted knowledge spillover effects. Table 9 of columns (1) and (3) show that DT improves enterprises' number of patent citations. DT promotes knowledge spillovers. Moreover, the coefficient of patent net citations is higher than the sum of citations. DT reduces the cost of acquiring knowledge for enterprises and provides a convenient way for them to acquire various knowledge

[1] We assign weights of 1, 2, and 3 to utility model patents, design patents, and invention patents, respectively.

Table 5
Non-linear analysis and robustness tests.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	*Patent*1	*Award*1	*Award*2	*RD*1	*Patent*1	*Patent*2	*Patent*1	*Patent*2
DCG_index	0.0954***	0.0220***	0.0215***	0.0004***				
	(11.6845)	(15.0070)	(15.0702)	(14.7622)				
DCG_index_squ	−0.0006***							
	(-6.0664)							
DCG_invest					0.0145***	0.0161***		
					(5.9248)	(5.9129)		
Digital R&D							0.0310***	0.0334***
							(14.8432)	(14.6618)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,488	10,488	10,488	10,488	10,488	10,488	10,488	10,488
R-squared	0.3739	0.5320	0.4574	0.3691	0.5123	0.4965	0.5165	0.5000

Note: Robust t-statistics are in parentheses. ***, ** and * indicate statistically significant at 1 %, 5 % and 10 % respectively. Fixed effects include city, year, and firm fixed effects. Control variables and parameter items are no longer reported to save space.

Table 6
Endogeneity test.

Variables	(1)	(2)	(3)
	First-stage	Two-stage	Two-stage
	*DCG*1	*Patent*1	*Patent*2
IV	−0.2010***		
	(-8.3500)		
DCG_index		0.1753***	0.1996***
		(7.8382)	(7.8001)
Controls	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes
Observations	11,841	11,841	11,841
Cragg-Donald Wald F statistic	69.73***		

Note: Robust t-statistics are in parentheses. ***, ** and * indicate statistically significant at 1 %, 5 % and 10 % respectively. Fixed effects include city, year, and firm fixed effects. Control variables and parameter items are no longer reported to save space.

materials. Columns (2) and (4) show that knowledge spillover promotes the ETI. We add *Sum_cite* and *Net_cite* to the model (4) and obtain the results for columns (5) and (6), respectively. Similarly, knowledge spillovers not only reinforce the positive effect of DT on ETI, but also strengthen the nonlinear relationship between the two. Knowledge spillovers have externalities, and with the support of new digital media, they can reduce the cost of knowledge acquisition for enterprise innovation. Thus, we can respond to hypothesis H2c that DT can improve the ETI by promoting knowledge spillovers.

Through the above analysis, we find that the scale of demand, cooperative innovation, and knowledge spillover are important mechanisms. They not only enhance the positive effect of DT on ETI, but also strengthen the inverted-U-shaped effect of DT on of ETI.

4.6. Further analysis

The DT of enterprises cannot be separated from the support of digital infrastructure (DI). Well-established and advanced DI reduces costs for DT and ETI. Previous studies have analyzed the relationship between DI and innovation. Osei (Osei, 2024) found a positive relationship between DI and innovation in Africa. Tian and Lu (Tian & Lu, 2023) found that China's DI promotes cross-regional innovation among enterprises. DI can also accelerate the flow of innovation resources and induce the growth effect of innovation. The above literature focuses on the impact of DI on innovation, but it is unclear what role of DI plays in the mechanism by which DT influences ETI. Henfridsson and Bygstad (Henfridsson & Bygstad, 2013) analyzed the drivers of the evolution of DI and concluded that adoption, innovation, and expansion were the main factors. The development of China's DI is largely driven by government policies. China' urbanization and industrialization demonstration zone city policy has promoted the innovation of digital technology of enterprises (Feng & Nie, 2024). We validate hypothesis **H3** to clarify the impact of regional DI on the DT and ETI.

Table 7
Regression results of demand scale effect.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
	OEprofit	*Patent*2	*Client*	*Patent*2	*Patent*2	*Patent*2
DCG_index	0.0572***	0.0111***	0.0350***	0.0272***	0.0662***	0.1262***
	(2.9485)	(4.4210)	(3.9325)	(8.7095)	(6.4224)	(5.9660)
OEprofit		0.0030**			0.0162***	
		(2.4951)			(5.2096)	
Client				0.1010**		1.8092***
				(2.1401)		(3.2184)
DCG_index_squ					−0.0003**	−0.0008***
					(-2.4398)	(-3.2069)
DCG_M					0.0004***	−0.0818***
					(6.8911)	(-3.0356)
DCG_squ_M					0.0001***	0.0009***
					(4.8730)	(3.0180)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9,612	9,318	3,284	3,284	9,612	3,306
R-squared	0.3532	0.8209	0.2431	0.6650	0.4123	0.4595

Note: Robust t-statistics are in parentheses. ***, ** and * indicate statistically significant at 1 %, 5 % and 10 % respectively. Fixed effects include city, year, and firm fixed effects. Control variables and parameter items are no longer reported to save space.

Table 8
Regression results of collaborative innovation mechanism.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
	Cooperation	*Patent2*	*Co_Patent1*	*Co_Patent2*	*Co_Patent3*	*Patent2*
DCG_index	0.4605***	0.0245***	0.0132***	0.0167***	0.0083***	0.1166***
	(14.3780)	(13.9825)	(9.1972)	(9.3573)	(7.9607)	(7.1628)
Cooperation		0.0017***				0.0236***
		(3.3390)				(3.4709)
DCG_index_squ						−0.0008***
						(-4.3832)
DCG_M						−0.0011***
						(-3.3954)
DCG_squ_M						0.0001***
						(3.2911)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,442	10,442	10,442	10,442	10,442	10,442
R-squared	0.2944	0.5043	0.3603	0.3598	0.3154	0.4537

Note: Robust t-statistics are in parentheses. ***, ** and * indicate statistically significant at 1 %, 5 % and 10 % respectively. Fixed effects include city, year, and firm fixed effects. Control variables and parameter items are no longer reported to save space.

Table 9
Regression results of knowledge spillover mechanism.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
	Sum_cite	*Patent2*	*Net_cite*	*Patent2*	*Patent2*	*Patent2*
DCG_index	0.0394***	0.0034**	0.0399***	0.0032*	0.1694***	0.1698***
	(24.4002)	(2.0588)	(24.7795)	(1.9560)	(8.7309)	(8.7189)
Sum_cite		0.4881***			0.8405***	
		(38.5002)			(8.8234)	
Net_cite				0.4851***		0.8432***
				(37.9111)		(8.7970)
DCG_index_squ					−0.0019***	−0.0020***
					(-8.2102)	(-8.2112)
DCG_M					−0.0201***	−0.0204***
					(-4.4605)	(-4.4980)
DCG_squ_M					0.0003***	0.0003***
					(5.2571)	(5.2974)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9,368	9,368	9,368	9,368	9,368	9,368
R-squared	0.5729	0.5857	0.5751	0.5840	0.5448	0.5430

Note: Robust t-statistics are in parentheses. ***, ** and * indicate statistically significant at 1 %, 5 % and 10 % respectively. Fixed effects include city, year, and firm fixed effects. Control variables and parameter items are no longer reported to save space.

We adopt the following strategy. First, the data released by Tsinghua University on China's New Infrastructure Competitiveness Index[2] supports our research. The index constructs first-level indicators from three dimensions: information infrastructure, convergence infrastructure and innovation infrastructure. The secondary indicators include 12 aspects, such as: communication network infrastructure, new technology infrastructure, computing infrastructure, industrial Internet, smart energy infrastructure, industrial technology innovation infrastructure, etc. They divide all provinces into four gradients according to their scores. We put the provinces in the first and second tiers into the higher group of DI, and the third and fourth tiers are in the lower group. We set the variable *Infras*. *Infras*=1 if the firm is in the high-level group; Otherwise, it is 0. Second, we performed a regression of DT effects on ETI in groups. The results are shown in columns (1) and (2) of Table 10. Regardless of whether it is a high-level or low-level DI region, DT has a significant effect in promoting ETI. Moreover, the coefficient value was larger in the area with high levels of DI, and the test for coefficient difference between groups is significant at 5 %. This fully shows that the level of DI can strengthen the positive effect of DT on ETI. Thirdly, we further explore whether DI can affect the mechanism of DT on ETI. We performed regression in groups. Columns (3)–(8) show that the scale effect of demand and the innovation cooperation mechanism have played a significant role in the high-level DI region. However, it was not significant in areas with low levels of DI. The knowledge spillover mechanism is not sensitive to DI. The above regression results provide an explanation for the slow progress of DT and the inefficiency of technological innovation in some regions of China.

Finally, we respond to the **H3**. First, well-developed DI could strengthen the promoting effect of DT on ETI. The more developed the DI, the greater the reinforcement effect. Second, high-level DI promotes the scale effect of demand and innovation cooperation mechanism, and the knowledge spillover mechanism is not sensitive to the development degree of DI. Therefore, improving regional DI can boost DT and ETI behaviors of micro enterprises.

5. Conclusions, limitations and policy implications

5.1. Conclusions

This study empirically examines the effects and mechanisms of Digital Transformation (DT) on Enterprise Technological Innovation (ETI) using data from manufacturing companies listed on the Shanghai and Shenzhen A-shares. Employing a panel fixed-effect model, the results reveal that DT exerts a significant positive impact on ETI. However, this relationship is non-linear, exhibiting an inverted U-shaped pattern.

[2] Detailed information can be seen at https://www.iii.tsinghua.edu.cn.

Table 10
Role of digital infrastructure.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	High level	*Low level*	*Patent2*	*Patent2*	*Patent2*	*Patent2*	*Patent2*	*Patent2*
DCG_index	0.0243***	0.0187***	0.0409***	0.0142**	0.0234***	0.0189***	0.0036**	0.0014
	(12.9431)	(3.5349)	(24.1346)	(2.3168)	(12.4405)	(3.4952)	(1.9907)	(0.2769)
OEprofit			0.0027**	−0.0032				
			(2.0419)	(-1.6369)				
Cooperation					0.0021***	−0.0003		
					(3.6302)	(-0.2154)		
Sum_cite							0.4518***	0.4900***
							(32.6656)	(14.0062)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,802	1,924	6,878	1,703	7,776	1,918	6,993	1,692
R-squared	0.4959	0.5916	0.4380	0.5894	0.4960	0.5911	0.5709	0.6496
Chow-test	P = 0.046		P = 0.000		P = 0.013		P = 0.000	

Note: Columns (2)–(5) are the regression results of the DID model. Columns (3), (5), and (7) are the groups with higher levels of digital infrastructure. Columns (4), (6), and (8) are the groups with lower levels of digital infrastructure. The fixed effect includes year and city fixed effects. Robust t-statistics are in parentheses. ***, ** and * indicate statistically significant at 1 %, 5 % and 10 % respectively. Control variables and parameter items are no longer reported to save space. The sample size of group regression is different, and the coefficients cannot be directly compared. We used the Chow-test method to investigate the significance of inter group coefficient differences. The above P-values indicate that the difference in coefficients between groups is statistically significant at 5 %, 1 %, 5 % and 1 % respectively.

Specifically, the positive effect of DT on ETI improves significantly within an optimal range, but beyond this threshold, the promotional effect diminishes considerably. Nonlinear relationships have also been validated in the mechanism of DT's impact on ETI. This finding underscores the importance for enterprises to carefully manage their DT investments to maximize the benefits for ETI while avoiding resource inefficiencies.

Furthermore, this paper delves into the mechanisms through which DT influences ETI. Grounded in demand-pull theory, open innovation theory, and knowledge spillover theory, the study identifies three key pathways: DT stimulates market demand, fosters cooperative innovation, and facilitates knowledge spillover, all of which collectively enhance ETI. Additionally, the research highlights the critical role of DI in supporting DT and its subsequent impact on ETI. The analysis demonstrates that high-level DI significantly amplifies the positive effect of DT on ETI, operating through the aforementioned mechanisms. This finding emphasizes the importance of regional DI development and suggests that policymakers should prioritize strategies to address regional disparities in DI, thereby creating an enabling environment for DT and ETI advancement. In conclusion, this study provides valuable insights into the nuanced relationship between DT and ETI, emphasizing the need for strategic DT investment and robust DI development to foster technological innovation in enterprises.

5.2. Limitations and discussion

Although our study offers valuable insights into the role of DT in fostering innovation, several limitations should be acknowledged. First, the measurement of DT relies on an exponential construction method, as direct data representing the level of DT in enterprises is unavailable. We constructed an index system encompassing digital strategy, digital investments, digital R&D, and digital technology applications to comprehensively capture various dimensions of DT. However, DT success also depends on intangible factors such as organizational culture, which cannot be directly quantified. While organizational culture is not explicitly measured, its influence may be indirectly reflected in metrics like digital strategy and investment. Our conclusions are supported by robustness tests using secondary indicators to measure DT, suggesting that measurement limitations have minimal impact on the overall findings.

Second, our analysis confirms an inverted U-shaped relationship between DT and ETI based on data from Chinese enterprises. However, the generalizability of this finding to other countries requires further investigation. The level of DI varies significantly across countries, influencing the costs and outcomes of DT. For instance, countries with broader DI coverage tend to experience lower DT costs for enterprises (Bharadwaj et al., 2013). We recommend extending this nonlinear analysis to both developed and developing economies, conducting comparative studies to enhance the external validity of the results.

Additionally, our study focuses on the manufacturing sector, leaving the impact of DT on ETI in the service sector unexplored. The goals and pathways of DT differ substantially between manufacturing and services. Manufacturing DT prioritizes production efficiency and supply chain optimization, typically starting with production process digitization. In contrast, service sector DT emphasizes enhancing customer experience and often begins with client-facing digitization. Future research should examine whether the observed relationship between DT and ETI holds in the service sector, which would broaden the scope of our findings and provide policymakers with industry-specific insights for promoting DT across diverse sectors. These limitations highlight opportunities for further research to deepen the understanding of DT's role in innovation and its contextual dependencies.

5.3. Policy implications

Based on the empirical findings, this study proposes the following policy recommendations to optimize DT and innovation:

First, enterprises should strategically manage initial DT investments to mitigate the crowding-out effect on innovation resources. Given the inverted U-shaped relationship between DT and innovation performance, firms should allocate resources between hardware (e.g., smart devices, sensors) and software (e.g., data analytics tools) based on their digital maturity. Prioritizing high-return projects, such as production automation or supply chain digitization, in the early stages can maximize efficiency. Additionally, investment ratios between data-driven elements and traditional factors (e.g., labor, capital) should be dynamically adjusted across DT phases. Establishing a robust evaluation mechanism to regularly review investment outcomes and adjust resource allocation is crucial to maintaining optimal DT investment levels.

Second, enhancing enterprises' digital capabilities and fostering collaborative innovation are essential. Governments should encourage firms to leverage open digital platforms (e.g., industrial Internet platforms, cloud computing) to access external resources and reduce R&D costs. Supporting the creation of regional digital collaboration platforms led by industry associations or leading firms can facilitate knowledge sharing and technical cooperation. These efforts can amplify network effects and promote early access to innovation ecosystems, benefiting a

broader range of enterprises.

Finally, governments should prioritize the development of digital infrastructure (DI) across regions, tailoring strategies to project types. Market-driven mechanisms should attract private investment for high-return projects such as 5G base stations and cloud computing centers. Public welfare projects, such as broadband networks in remote areas, require increased fiscal support to ensure implementation. For innovative but uncertain-return projects (e.g., blockchain infrastructure), public-private partnerships (PPPs) and pilot programs can be adopted to evaluate and refine financing models. Establishing a monitoring framework to track project progress is necessary to ensure efficient resource utilization and accountability.

Declaration of competing interest

It is declared this article has no conflicting interest.

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Data availability

Data will be made available on request.

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